

Phase 9 Closeout: Budget-Control CVRP Study

Purpose

This protocol defines the bounded closeout study attached to phase 9.

The goal is not to introduce a new benchmark family or reopen the broad model-strength question. The goal is to answer the narrow mechanism question left open by the bounded real-world CVRP study:

when the phase-9 CVRP pipeline improves over weaker baselines, how much of that gain is better explained by direct synthesis, by extra candidate budget, or by replay-aware iterative solver evolution?

The closeout therefore stays on the same bounded CVRPLIB X setup, the same validator, the same train/held-out split, and the same generator and judge models already used in the retained phase-9 evidence.

Relationship To Phase 9

This is a phase-9 closeout study, not a new broad phase in the sense of phases 7, 8, or 9.

- Phase 9 established that bounded whole-solver evolution on real-world CVRP is technically feasible and can beat weaker fixed baselines on held-out instances.
- Phase 9 did not show a win over the strongest fixed Clarke--Wright baseline on the primary endpoint.
- A historical descriptive direct Codex-authored CVRP check was stronger than the replicated autonomous mean. That result was not a frozen replicated condition and is not retained as thesis evidence.

The closeout study exists to convert that descriptive gap into a controlled budget-comparison result, or else to falsify the stronger interpretation cleanly.

Naming And Archival Structure

The branch and tag naming follow the same project structure used for the earlier phases.

- Working branch: `cvrp-closeout-budget-control`
- Official archive tag on successful completion: `phase-9-closeout-budget-control`
- Smoke outputs: DO NOT COMMIT/`phase9_closeout_budget_control_smoke/`
- Official outputs: `runs/cvrp_phase9_closeout/budget_control/`
- Permanent report: `docs/PHASE_9_CLOSEOUT_BUDGET_CONTROL_CVRP_RESULTS_YYYY-MM-DD.md`

The official tag should be created only after:

1. the protocol is frozen,
2. the smoke campaign passes,
3. the official replicated campaign completes, and
4. the aggregate report and thesis integration are checked.

Fixed Invariants

The following parts of the study are fixed unless this protocol is revised before any official run:

- Benchmark family: the same bounded phase-9 CVRPLIB X suite
- Train/held-out split: unchanged from phase 9
- Generator model: unchanged from phase 9
- Judge model: unchanged from phase 9
- Timeout policy: unchanged from phase 9 unless a technical-failure audit justifies a documented change
- Primary endpoint: held-out penalized gap
- Secondary endpoints: held-out feasibility, held-out runtime, accepted-candidate count where applicable, code novelty, and solver complexity
- Replication design: paired seed offsets shared across compared conditions

This preserves comparability with the retained phase-9 thesis result and avoids reopening the separate model-strength question.

Conditions

The official closeout suite retains the three fixed phase-9 baselines for context:

- nearest_neighbor_constructive
- clarke_wright_savings
- regret_insertion_local_search

The closeout comparison set is:

1. direct_generate_plus_one_repair
2. budget_matched_no_replay
3. replay_solver_evolution

The scientific roles of the closeout arms are:

- direct_generate_plus_one_repair: a bounded direct-synthesis control with one scripted evaluation-driven follow-up round
- budget_matched_no_replay: an independent multi-candidate control with the same candidate budget as iterative search but no replay memory
- replay_solver_evolution: an incumbent-preserving iterative search arm that receives compact replay summaries from earlier weak candidates

Candidate Budgets

The closeout uses the same 8-candidate budget as the bounded phase-9 evolutionary run:

- direct_generate_plus_one_repair: 2 generated candidates
- budget_matched_no_replay: 8 independent generated candidates
- replay_solver_evolution: 8 iterative generated candidates

The direct arm uses a smaller budget because it asks a different question: whether minimal bounded repair already explains most of the descriptive direct-solver advantage.

Statistical Reporting

The official report follows the same conservative reporting discipline as the retained thesis phases.

- Report per-condition held-out means on the primary endpoint.
- Report paired deltas with 95% bootstrap confidence intervals.
- For the closeout arms, accept or retain candidates using train-panel feasibility first and train penalized gap second; report runtime, but do not use runtime as the primary acceptance criterion.
- Report technical-validity checks, including generation errors, fallback counts, feasibility failures, and accepted-candidate counts.
- If API budgeting matters, record spend separately from provider usage or billing logs; the repository's built-in closeout reports do not currently estimate dollar cost.
- Keep smoke outputs out of the thesis evidence base.
- Do not promote replay-specific claims unless `replay_solver_evolution` beats `budget_matched_no_replay`.

Success Criterion

The strongest positive closeout result is not necessarily "beat Clarke--Wright." The closeout is scientifically successful if it cleanly answers the budget-versus-mechanism question under the phase-9 validator.

Valid closeout outcomes include:

- replay-aware evolution beats budget-matched no replay under paired uncertainty,
- budget-matched no replay matches replay and shows that extra search budget explains most gains,
- direct synthesis plus one repair dominates both iterative arms, which narrows the thesis conclusion toward bounded coding assistance rather than replay-aware autonomy.

Any of these outcomes is more valuable than another unconstrained late benchmark extension because each sharpens the final dissertation claim without broadening it.